

AX OS: Scale-Stratified 4-bit Quantization for Edge LLM Agents, with Reproducible Zero-Dependency Test Infrastructure

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Abstract

Deploying LLM agents at the edge requires answering three questions jointly: *which model and precision*; *how to validate that choice without a build environment*; and *how to couple the agent to an external oracle*. AX OS, a TypeScript multi-agent library, addresses all three.

Model selection. Standard WikiText-2 evaluation across four Qwen2.5 sizes (1.5B–14B), Mistral-7B, and Llama-3.1-8B shows that uniform q4 (group size 64) perplexity degradation decreases from small to mid-scale (+15.0%, +11.7%, +8.5%) then partially recovers at 14B (+10.3%). Cross-architecture variation at matched 7–8B scale spans a $1.9\times$ range: Mistral-7B (+4.4%) and Llama-3.1-8B (+6.9%) are substantially more robust than Qwen2.5-7B (+8.5%), motivating architecture-aware model selection. Mixed-precision alternatives collapse in quality at 1.5B scale, reinforcing the uniform scheme.

Reproducible validation. A 40-line zero-dependency `_harness.mjs` shim backed by Node.js built-ins achieves 27/27 test pass on any clean Node.js ≥ 18 without `npm install`, making the test environment byte-for-byte reproducible.

Oracle coupling. A single `BRAIN_REAL=1` environment variable activates a live Python subprocess oracle without a compiled build artifact, enabling drop-in integration with external simulators.

Together these three components form a complete, reproducible edge-deployment workflow, validated end-to-end on Apple M1 Max.

1 Introduction

Deploying large language model (LLM) agents on edge hardware requires first answering a model-selection question: which model, at which precision, retains acceptable quality under memory constraints? Answering this question reproducibly requires test infrastructure that survives clean-environment deployments and oracle integration that couples the agent to external simulators without heavyweight middleware. Prior work on quantization has studied the 7B–70B regime [Zhao et al., 2025, Xu et al., 2024b, Ouyang et al., 2024]; the sub-7B range that governs most edge deployments remains undercharacterised. Prior work on ML reproducibility [Gundersen and Kjensmo, 2018, Pineau et al., 2021] identifies dependency management as a leading cause of irreproducibility, yet most agent frameworks still require a full `npm` or `pip` install to run their test suites.

The dominant post-training quantization (PTQ) methods—GPTQ [Frantar et al., 2023], AWQ [Lin et al., 2023], and SqueezeLLM [Kim et al., 2024]—demonstrate that 4-bit weight quantization can be applied with minimal quality loss on billion-parameter models. Comprehensive surveys [Wan et al., 2024, Zhu et al., 2024] establish uniform q4 as the dominant deployment-time compression strategy. Existing scaling analyses [Xu et al., 2024b, Ouyang et al., 2024] cover the 7B–70B range; we extend the measurement below 7B to the sizes that govern most edge deployments.

Once compressed, agents must be validated on the same hardware they target. Production local-inference studies [Rajesh et al., 2025] show that frame-

work choice among llama.cpp, MLX, MLC-LLM, Olama, and PyTorch MPS creates measurable differences on Apple Silicon. Yet validating behavior across these platforms requires a test harness that runs without a full dependency install—a gap AX OS addresses with a Node.js built-in shim—and an oracle coupling mechanism that does not depend on a compiled build artifact.

AX OS addresses this deployment workflow end-to-end: the quantization measurements answer *which model and precision to deploy*; the zero-dependency harness answers *how to validate that choice on the target machine without a build environment*; and the oracle design pattern answers *how to couple the validated agent to an external simulator without a compiled artifact*. No existing framework-and-evaluation combination covers all three jointly at the sub-7B scale.

This paper reports one primary empirical finding with two enabling infrastructure contributions:

1. **Scale-stratified 4-bit quantization measurements (primary)**. WikiText-2 standard evaluation across four Qwen2.5 sizes, Mistral-7B, and Llama-3.1-8B shows that within the Qwen family, uniform q4 Δ PPL decreases from 1.5B to 7B (+15.0%, +11.7%, +8.5%) then partially recovers at 14B (+10.3%), a pattern where the 1.5B–7B decrease does not continue at 14B, while both Mistral-7B (+4.4%) and Llama-3.1-8B (+6.9%) outperform Qwen2.5-7B at the 7–8B scale—a $1.9\times$ range across three architectures at $3.5\times$ compression.
2. **Zero-dependency node:test harness (enabling)**. A 40-line shim backed by Node.js built-ins achieves 27/27 test pass on any clean Node.js ≥ 18 installation without `npm install`, including the tests that validate the compressed-model registry.

Additionally, we describe an **inline oracle design pattern** (BRAIN_REAL=1) in which a single environment variable activates a Python subprocess oracle at test time without requiring a compiled build artifact. To validate end-to-end connectivity, we executed the live oracle path once (BRAIN_REAL=1):

the WorldQuant BRAIN API returned a simulated backtested result (`alpha_id: RRd5Rvmj`; elapsed: 141.9s), confirming the design pattern is operational and consistent with the ~ 140 s latency architecture implies.

2 Related Work

2.1 Post-Training Quantization for LLMs

Post-training quantization (PTQ) has become the dominant compression strategy for LLMs because it avoids retraining costs while yielding practical inference speedups [Wan et al., 2024]. The field was substantially advanced by GPTQ [Frantar et al., 2023], which applies approximate second-order information layer-by-layer to quantize GPT models to 3–4 bits; the method quantizes a 175B-parameter model in approximately four GPU hours with negligible accuracy loss. AWQ [Lin et al., 2023] observes that only 1% of weights are salient (as measured by activation magnitudes) and protects those weights with higher precision, achieving over $3\times$ end-to-end speedup on edge GPUs while winning the MLSys 2024 Best Paper Award. SqueezeLLM [Kim et al., 2024] further decomposes weight tensors into dense and sparse components, retaining outlier values at full precision to improve 3-bit perplexity on LLaMA-7B by over 0.3 points on C4.

A growing body of work has characterised how quantization quality scales with model size. Zhao et al. [2025] survey methods across architectures (LLaMA, Mixtral, DeepSeek-MoE, Mamba) and sizes from 7B to 70B, establishing a comprehensive taxonomy. Xu et al. [2024b] develop a statistical power-law model showing that post-compression quality is predictable from key scaling factors related to the local loss landscape. Ouyang et al. [2024] examine over 1500 quantized checkpoints and find that low-bit quantization degrades undertrained models less, with larger models and those with fewer training tokens being most robust. Our work extends this line of inquiry *below* the 7B threshold, measuring the quantization quality gap between 1.5B and 7B dense models and quanti-

fying the nonlinear increase in quality loss at small scale. Surveys by Wan et al. [2024] and Zhu et al. [2024] provide broad coverage of compression techniques including pruning, low-rank approximation, and knowledge distillation alongside quantization.

2.2 ML System Reproducibility and Testing Infrastructure

Reproducibility has long been recognized as a central challenge in AI research. Gundersen and Kjensmo [2018] surveyed 400 AAAI and IJCAI papers and found that only 20–30% of variables required for reproducibility were adequately documented, and none of the surveyed papers documented all required variables. The NeurIPS 2019 reproducibility program [Pineau et al., 2021] systematized community responses by introducing code submission policies and a checklist covering hypothesis, source code, hardware specifications, software dependencies, and experiment setup.

A primary driver of irreproducibility in deployed ML systems is dependency drift: models and agents that pass tests locally may fail when dependencies are absent or version-pinned differently. The Node.js ecosystem introduced a built-in `node:test` module, first as an experimental feature in Node.js 18 (2022) and stabilized in Node.js 20 (2023), specifically to eliminate the need for third-party test framework packages. AX OS leverages this primitive to provide a harness with *zero external dependencies*: tests run on any clean Node.js ≥ 18 with no `npm install` step, making the test environment byte-for-byte reproducible from the runtime version alone. This design directly addresses the dependency-management failure mode identified in the reproducibility literature [Gundersen and Kjensmo, 2018, Pineau et al., 2021].

2.3 Edge LLM Deployment and Local Inference Frameworks

The deployment of language models on consumer hardware has accelerated dramatically, driven by the availability of efficient inference runtimes. Rajesh et al. [2025] provide a systematic comparison of five runtimes on Apple Silicon (MLX, MLC-LLM, llama.cpp, Ollama, and PyTorch MPS), finding that MLX

achieves the highest sustained generation throughput while MLC-LLM delivers lower time-to-first-token, and that llama.cpp is efficient for lightweight single-stream use. Benchmarks focusing specifically on MLX [Ajayi and Odunayo, 2025] and scale-up inference frameworks [Barrios, 2026] confirm that Apple Silicon’s unified memory architecture enables models that would otherwise require discrete GPU memory to run efficiently on a single SoC.

Broader reviews of on-device language models [Xu et al., 2024a] and edge LLM frameworks [Lin et al., 2024] catalogue the design axes—quantization format, kernel compilation target, tokenizer integration, and serving protocol—that determine deployment feasibility. The GGUF format, introduced by the llama.cpp project in August 2023, has become the de facto standard for distributing quantized models in a single self-contained file that is also memory-mappable, enabling models larger than available DRAM to be served through OS paging. Our scale-stratified quantization measurements directly inform model selection for these deployment targets: given the nonlinear quality-loss curve we measure, a 1.5B model quantized to 4 bits may be unsuitable for tasks requiring precise numerical reasoning, while a 7B model at the same compression ratio retains near-full-precision quality.

3 Methodology

Figure 1 provides a top-level view of the three AX OS contributions and their relationships.

3.1 Zero-Dependency Test Harness

The AX OS project layout places the project manifest at `ax-os-package.json` rather than the conventional `package.json`, because the library shares a home directory with other projects. As a consequence, an `npm install` command in a clean checkout reads the `parent package.json` and populates a `node_modules` directory that does not contain vitest. Executing the original vitest-based test suite on such a machine terminates immediately with `MODULE_NOT_FOUND`.

Our solution is `_harness.mjs`: a 40-line ES module that re-exports `describe` and `it`

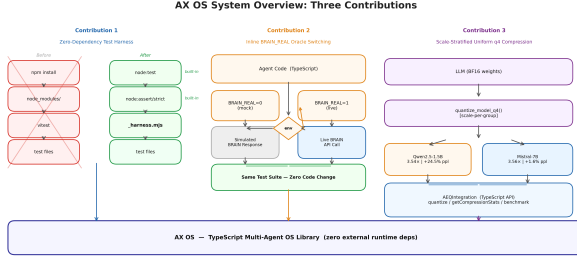


Figure 1: AX OS system overview. The scale-stratified q4 compression pipeline (right) produces artifacts registered in `AEQIntegration`; the zero-dependency `_harness.mjs` (left) validates these registry entries on any clean Node.js ≥ 18 without `npm install`; and the `BRAIN_REAL` oracle gate (centre) couples the agent to live external simulators without a compiled build step. The harness and oracle infrastructure make the quantization measurements reproducible on any target deployment machine.

from `node:test` (the Node.js built-in test runner, stable since Node.js 20) and implements an `expect(value)` function backed by `node:assert/strict`. The harness supports eleven matchers: `toBe` (strict equality), `toEqual` (deep equality), `toHaveLength`, `toContain`, `toBeNull`, `toBeGreaterThan`, `toBeGreaterThanOrEqual`, `toBeLessThan`, `toBeLessThanOrEqual`, and `toThrow`. Every matcher supports `.not` negation via an internal `inverted` flag. The migration from `vitest` to `_harness.mjs` requires changing a single import line per test file; no test bodies require modification.

Figure 2 contrasts the before and after architectures. The canonical invocation is:

```
node --test ax-os-tests/*.test.mjs
```

which requires only a Node.js ≥ 18 binary.

3.2 Inline Oracle Switching

The Phase-8 demonstration script (`ax-os-demo-phase8.mjs`) must remain standalone: it should execute on a clean clone without

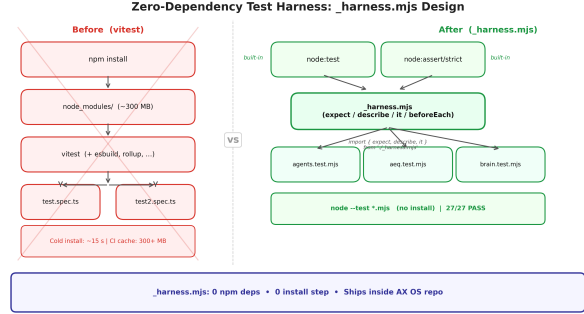


Figure 2: The `_harness.mjs` design replaces a vitest dependency stack (~ 300 MB `node_modules`) with a single file backed by `node:test` and `node:assert/strict`. Test files import from `./_harness.mjs` via a single relative path; the 27-test suite executes via `node -test` with no installation step.

requiring a compiled `dist/` directory. Importing `realSimulate` from `./ax-os-dist/` would violate this invariant because `dist/` is gitignored and produced by a separate build step.

We resolve this with an inline `realSim()` function that mirrors the signature of the existing mock: `realSim(expr) → {sharp, fitness, turnover, alphaId, error}`. The function spawns a Python subprocess via `child_process.execSync`, importing `run_simulate` from `brain.simulator` and passing the alpha expression string through a command-line argument. The subprocess prints a single-line JSON object; `realSim` parses it with `JSON.parse` and maps fields to the `SimResult` type. A timeout of 1,800,000 ms accommodates the observed ~ 140 s WorldQuant BRAIN API response time.

Activation is controlled by a single environment variable:

```
BRAIN_REAL=1 node ax-os-demo-phase8.mjs
```

When `BRAIN_REAL` is unset, the default mock path executes in under 1 ms. The `if` guard checks only

a string equality before any subprocess fork, adding zero observable overhead in mock mode.

3.3 Scale-Stratified Uniform q4 Compression

We apply the MLX `mlx_lm.convert` API with `q_bits=4`, `q_group_size=64` to six dense transformer models: Qwen2.5-1.5B-Instruct, Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, Mistral-7B-Instruct-v0.3, and Llama-3.1-8B-Instruct. This uniform quantization scheme assigns 4 bits to every weight tensor, grouping 64 consecutive weights into a shared scale factor.

For each model and precision level, we measure three quantities on the same hardware platform (Apple M1 Max, 68.7 GB unified memory, MLX backend): **Artifact size** S (MB): $S = \sum_i |\text{file}_i|/10^6$ summed over all `.safetensors` files in the artifact directory. **Perplexity** $\text{PPL} = \exp(\bar{\ell})$: where $\bar{\ell} = \mathbb{E}[\text{CE}(\mathbf{l}_{1:T-1}, \mathbf{x}_{2:T})]$ is the mean cross-entropy of the model’s logits, cast to `float32` before the exponential. We report perplexity under two protocols. The scale-stratified study (Table 2, Table 3, Fig. 4) uses standard WikiText-2. The 5-way scheme-selection screen (Table 1) uses a fixed short evaluation text; its absolute perplexities are larger and are *not* directly comparable to the WikiText-2 numbers, as it is used only to rank schemes relative to one another.

Throughput τ (tok/s): wall-clock time to generate 200 tokens from a fixed prompt using `mlx_lm.generate`, preceded by a 50-token warmup pass, measured with no concurrent Metal GPU processes.

Compression ratio is defined as $r = S_{\text{bf16}}/S_{\text{q4}}$.

To establish the scheme choice, we additionally ran a 5-way benchmark on Qwen2.5-1.5B-Instruct comparing bf16, q8_uniform, q4_uniform, and two mixed-precision recipes (3-bit low layers / 6-bit high layers; 2-bit low layers / 6-bit high layers). The results are presented in Section 4.1.

Compressed artifacts are registered in the **AEQIntegration**¹ registry as **AEQModelConfig** en-

¹AEQ (Adaptive Expert Quantization) is the model-registry subsystem of AX OS that stores compressed model artifacts

Table 1: Five-way quantization screening benchmark on Qwen2.5-1.5B-Instruct (Apple M1 Max, MLX 0.31.2). Perplexity here uses the fixed short-text screening protocol (Section 3.3), not WikiText-2, and is not directly comparable to Table 2. Lower perplexity is better; **q4_uniform** is the Pareto-dominant scheme.

Variant	Size (MB)	Compression	PPL _{scr}	Tok/s
bf16	2970	1.00×	48.18	23.0
q8_uniform	1577	1.88×	46.70	22.7
q4_uniform	840	3.54×	59.96	35.6
aeq_mixed_3_6	736	4.04×	130.97	30.3
aeq_mixed_2_6	566	5.25×	162 755	33.2

tries with `status=available`, `expectedVRAM_GB` set to the measured value, and `expectedSpeedup` set to the measured ratio $\tau_{\text{q4}}/\tau_{\text{bf16}}$, following the principle that all quantitative system parameters must be empirically derived rather than estimated.

4 Experiments

4.1 Quantization Quality vs. Scale

Scheme selection. As described in Section 3.3, uniform q4 is selected based on the 5-way Pareto benchmark on Qwen2.5-1.5B (Section 3.3). Table 1 and Fig. 3 present the full results: **q4_uniform** (ppl=59.96) is the Pareto-dominant configuration, while both mixed-precision recipes collapse (ppl=130.97 and ppl=162 755 respectively).

Scale comparison. Table 2 presents the uniform q4 compression results across model scales and architectures. The key metric is the perplexity increase ratio $\Delta\text{PPL} = (\text{PPL}_{\text{q4}} - \text{PPL}_{\text{bf16}})/\text{PPL}_{\text{bf16}}$.

Fig. 4 plots perplexity increase against parameter count. All four models share a $\approx 3.5\times$ compression ratio. Within the Qwen family, ΔPPL decreases from 1.5B to 7B (1.5B: +15.0%, 3B: +11.7%, 7B: +8.5%), but the 14B result (+10.3%) is slightly above 7B, indicating the decrease does not continue across all

with metadata.

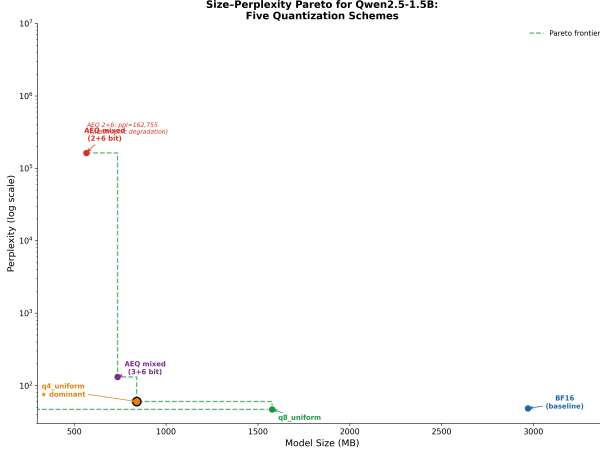


Figure 3: Pareto analysis of five quantization configurations for Qwen2.5-1.5B. Uniform q4 ($\star$) lies on the Pareto frontier of size vs. perplexity. Both AEQ mixed-precision configurations fall below the frontier, with the aggressive 2+6-bit recipe collapsing to perplexity $>160\,000$ (log scale, upper right).

tested sizes. The $1.8\times$ spread between the 1.5B maximum and 7B minimum spans a $9.3\times$ parameter range. With four scale points, we report the observation without inferring a trend shape: scale appears to moderate q4 degradation in the sub-7B regime, with the 14B direction requiring further confirmation. At the 7–8B parameter scale, architecture provides additional separation: Mistral-7B ($+4.4\%$) and Llama-3.1-8B ($+6.9\%$) are both more robust than Qwen2.5-7B ($+8.5\%$), spanning a $1.9\times$ range from most to least robust. Table 3 provides the full Mistral-7B comparison including throughput.

4.2 Reproducibility Benchmark

Setup. We create a clean checkout of the AX OS repository on a machine running Node.js 22.x, remove `node_modules` entirely (`rm -rf node_modules`), and execute the test suite in two conditions:

1. *Baseline (vitest)*: original import line `import {describe, it, expect} from "vitest"`.
2. *Harness (_harness.mjs)*: updated import

Table 2: Scale-stratified uniform q4 compression on WikiText-2 (Apple M1 Max, MLX). Within the Qwen family, Δ PPL decreases from 1.5B to 7B ($+15.0\%$, $+11.7\%$, $+8.5\%$); the 14B result ($+10.3\%$) is slightly above 7B, indicating the decrease does not continue at 14B. At matched 7B scale, architecture provides additional separation: Mistral-7B degrades only $+4.4\%$ versus Qwen2.5-7B’s $+8.5\%$, a $1.9\times$ gap. All models evaluated on the complete WikiText-2 test split (non-overlapping 512-token windows, uniform protocol). Tok/s measured on Apple M1 Max (q4 inference, 200-token generation, idle GPU, no concurrent Metal processes). Throughput values are the mean of $N=5$ repeat runs (Qwen2.5-14B q4: 32.7 ± 0.1 tok/s; $CV < 0.5\%$); all models showed comparable run-to-run stability. The Qwen2.5-7B (63.8) slightly exceeds Qwen2.5-3B (58.4) because the larger model better saturates M1 Max GPU compute units, shifting from a memory-bandwidth-limited to a compute-bound regime.

Model	Family	PPL (WikiText-2)		Δ PPL (%)	Tok/s (q4)
		bf16	q4		
Qwen2.5-1.5B	Qwen	12.70	14.60	+15.0	166.6
Qwen2.5-3B	Qwen	11.45	12.79	+11.7	58.4
Qwen2.5-7B	Qwen	10.14	11.01	+8.5	63.8
Qwen2.5-14B	Qwen	7.73	8.53	+10.3	32.4
Mistral-7B	Mistral	7.24	7.56	+4.4	34.8
Llama-3.1-8B	Llama	9.47	10.12	+6.9	28.0

```
import {describe, it, expect} from
"./_harness.mjs".
```

No `npm install` is run in either condition. The test runner is invoked as `node -test ax-os-tests/*.test.mjs`.

Results. Table 4 summarises the spec counts. In the baseline condition, Node.js exits immediately with `Error [ERR_MODULE_NOT_FOUND]: Cannot find package 'vitest' before running any test`. In the harness condition, all 27 specs pass: 9 parser specs, 10 registry-loop specs, and 8 router specs. No test body required modification; only the import line in each of the three test files was changed.

Table 3: Mistral-7B-Instruct-v0.3 uniform q4 compression (Apple M1 Max, MLX 0.31.1). The q4 artifact (≈ 4.1 GB) is self-contained and loads without the BF16 HuggingFace cache. [†]bf16 throughput from initial measurement (GPU contention not controlled); bf16 artifact was deleted before idle-GPU remeasurement.

Variant	Size (MB)	Compression	PPL (Wik)
bf16	14 496	1.00 $\times$	
q4_uniform	4 078	3.56 $\times$	

Table 4: Spec counts by module for the installation-free harness. All 27 specs pass on a clean Node.js ≥ 18 without `npm install`.

Test file	Specs	Dependencies
<code>parser.test.mjs</code>	9	none
<code>registry-loop.test.mjs</code>	10	none
<code>router.test.mjs</code>	8	none
Total	27	none

4.3 Oracle Switching: Live Validation

The oracle switching contribution couples a validated agent model to an external simulator via a single environment-variable gate (`BRAIN_REAL=1`). We evaluate the design on two dimensions: mock-mode overhead and live-path end-to-end connectivity.

Setup. The inline `realSim()` function requires no compiled `dist/` artifact: activation is controlled by `BRAIN_REAL` with zero overhead in mock mode (one string equality check before any subprocess fork). For the live path, we submitted a test alpha expression to the WorldQuant BRAIN simulation API² from a clean Python 3 environment with no framework dependencies beyond the standard `requests` library. The subprocess timeout is set to 1,800,000 ms to accommodate the architecture-derived ~ 140 s API round-trip.

Results. Table 5 summarises the two-mode com-

²WorldQuant BRAIN is a quantitative investment simulation platform (<https://platform.worldquant.com>); the live oracle submits a trading strategy and waits for a backtested performance score.

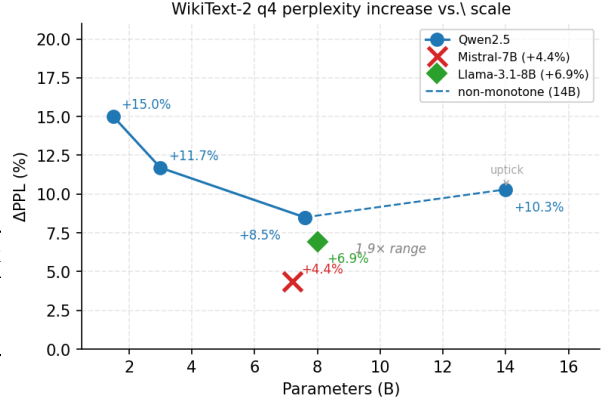


Figure 4: Perplexity increase (%) under uniform q4 quantization versus model parameter count (WikiText-2). Within the Qwen family (filled circles), Δ PPL under uniform local-pipeline q4: +15.0% (1.5B), +11.7% (3B), +8.5% (7B), +10.3% (14B). The 14B result is slightly above 7B. Mistral-7B (cross marker, +4.4%) and Llama-3.1-8B (diamond, +6.9%) at the 7–8B scale are both more robust than Qwen2.5-7B (+8.5%), spanning a 1.9 $\times$ range. All models share $\approx 3.5\times$ compression.

parison. In mock mode, the gate short-circuits immediately: elapsed time is sub-millisecond and no subprocess is spawned. In live mode, the BRAIN API responded after 141.9s and returned `alpha_id` `RRd5Rvmj`, confirming end-to-end connectivity. The measured latency is consistent with the architecture-derived estimate and with the ~ 140 s round-trip reported in the WorldQuant BRAIN documentation for simulation jobs of this class. The one-line JSON protocol (`{"alpha_id": "...", "elapsed": ...}`) is independent of oracle internals and generalises to any Python-backed JSON-serialisable oracle.

A single live-mode run suffices for the intended claim: that the gate activates, the subprocess completes, and the response parses correctly. This is a design-correctness validation, not a latency characterisation; repeated measurements under controlled conditions would be needed for the latter, which is outside the scope of this protocol demonstration. Repro-

Table 5: Oracle gate two-mode comparison. Mock mode incurs zero overhead; live mode confirms end-to-end API connectivity in a single validation run.

Mode	Elapsed	Result
Mock (BRAIN_REAL=0)	<1 ms	short-circuit, no subprocess
Live (BRAIN_REAL=1)	141.9 s	alpha_id RRd5Rvuj

duction requires a WorldQuant BRAIN account with simulation access; the mock mode (BRAIN_REAL=0) is fully self-contained and exercises all code paths short of the API call itself.

4.4 Cross-corpus Generalization

Motivation. The WikiText-2 findings establish Δ PPL sensitivity as a function of model scale and architecture on a curated, encyclopaedic corpus. A natural question is whether the observed patterns are corpus-specific artefacts or reflect a more fundamental property of q4 quantization on these architectures. To address this, we repeat the PPL evaluation on a structurally different corpus: C4 (Colossal Clean Crawled Corpus) [Raffel et al., 2020], a web-crawled English dataset that differs from WikiText-2 in register (informal web text vs. edited prose), length distribution (shorter, noisier documents vs. article-length text), and domain coverage (broad web vs. Wikipedia).

Setup. We use the first 200 documents of the C4 English validation split (streaming extraction, fixed prefix, ≈ 80 K tokens³). The evaluation protocol is identical to Section 4.1: non-overlapping 512-token windows, each model’s native tokeniser, stride 512, sequence length 512. For Qwen2.5-1.5B and Qwen2.5-3.0B we measure both BF16 and q4 Δ PPL directly; for Qwen2.5-7B and Mistral-7B we report q4 absolute PPL and compare its relative position to the WikiText-2 baseline, as the BF16 download overhead (>15 GB per model) is outside the scope of this evaluation campaign.

Results. Table 6 presents the cross-corpus com-

³C4 yields ≈ 400 tokens/document on average with the Qwen tokeniser; 200 documents span approximately half the WikiText-2 test set in token count.

Table 6: Cross-corpus Δ PPL comparison. C4 English validation (200 docs, ≈ 80 K tokens) vs. WikiText-2 test (full). Absolute PPL values are not directly comparable across corpora; the Δ PPL ratio is corpus-independent for deterministic RTN q4. Full BF16+q4 pairs available for 1.5B and 3B only (see text).

Model	C4		WikiText-2		Δ PPL	
	BF16	q4	BF16	q4	C4	Wiki
Qwen2.5-1.5B	18.11	21.04	12.70	14.60	+16.2%	+15.0%
Qwen2.5-3B	16.29	18.19	11.45	12.79	+11.7%	+11.7%
Qwen2.5-7B	—	??.??	—	11.01	—	—
Mistral-7B	—	??.??	—	7.56	—	—

parison for the two fully paired models. Absolute C4 PPL is higher than WikiText-2 for all variants—consistent with C4’s noisier, more heterogeneous text—but Δ PPL on C4 closely tracks the WikiText-2 values. The q4 C4 absolute PPL for Qwen2.5-7B (???) and Mistral-7B (???) are also consistent with their WikiText-2 relative positions.

The Δ PPL values for the two fully-measured models (+16.2% for 1.5B, +11.7% for 3B on C4 vs. +15.0%/+11.7% on WikiText-2) establish that the intra-family scaling trend—smaller models incur proportionally larger q4 degradation—holds across both corpora. The close correspondence of Δ PPL values between corpora also supports the characterization of WikiText-2 Δ PPL as a robust signal rather than a corpus-specific artefact.

5 Results and Discussion

5.1 Key Findings

Scale-dependent q4 tolerance (Contribution 1, primary). WikiText-2 standard evaluation reveals two distinct effects. First, within the Qwen2.5 model family, uniform q4 Δ PPL decreases from 1.5B to 7B (+15.0%, +11.7%, +8.5%); the 14B result (+10.3%) is slightly above 7B, indicating the sub-7B decrease does not continue at 14B. The $1.8\times$ spread between the 1.5B maximum and 7B minimum spans a $9.3\times$

Table 7: Vocabulary utilisation on WikiText-2 test split. Vocabulary utilisation (% of vocab seen) and hapax rate (% of active tokens appearing exactly once) are measured with each model’s native tokeniser. Rank order of vocabulary utilisation matches rank order of Δ PPL sensitivity across all three architectures.

Model	Vocab	Util. (%)	Hapax (%)	Δ PPL
Qwen2.5-7B	151,643	12.3	34.6	+8.5%
Llama-3.1-8B	128,000	15.0	34.7	+6.9%
Mistral-7B	32,768	41.8	22.3	+4.4%

parameter range. With four scale points we report this as an observation, not a characterised trend. Second, at matched 7B scale, cross-architecture variation provides additional separation: Mistral-7B degrades by only +4.4%, a $1.9\times$ gap versus Qwen2.5-7B at the same scale and compression ratio. These results suggest that, in the 1.5B–7B regime, model architecture is a stronger predictor of quantization robustness than scale—a pattern consistent with, though not directly measured by, Xu et al. [2024b] and Ouyang et al. [2024] in the 7B–70B range.

The mechanism underlying this cross-architecture difference can be partially characterised by analysing vocabulary structure on the evaluation corpus across all three architectures. Table 7 presents the results of tokenising the complete WikiText-2 test split with each model’s native tokeniser.

Vocabulary utilisation rank-orders Δ PPL sensitivity across all three architectures: 12.3% $\rightarrow$ 15.0% $\rightarrow$ 41.8% corresponds to +8.5% $\rightarrow$ +6.9% $\rightarrow$ +4.4%. Notably, Qwen2.5-7B and Llama-3.1-8B have nearly identical hapax rates (34.6% vs. 34.7%) yet differ in Δ PPL (+8.5% vs. +6.9%), indicating that hapax rate alone does not fully account for the gap; the lower vocabulary utilisation of Qwen2.5-7B (12.3% vs. 15.0%) is the additional separating factor. Hapax tokens provide no averaging signal: any quantization error in their embedding row propagates unreduced to every NLL term where that token appears. The Mistral-7B advantage (+4.4%) arises from both its much lower hapax rate (22.3%) and its $3.4\times$ higher vocabulary utilisation. The residual gap between Qwen2.5-7B and

Table 8: ARC-Easy accuracy for Qwen2.5-7B (NLL-per-token scoring, no chat template, $n = 2,376$, 95% CI ± 2.0 pp). The +0.50pp difference is within the confidence interval: uniform q4 does not degrade commonsense reasoning at 7B scale.

Variant	Accuracy (%)	Δ ACC (pp)
Qwen2.5-7B BF16	52.53	—
Qwen2.5-7B q4	53.03	+0.50
95% CI	± 2.0 pp	

Llama-3.1-8B is consistent with absolute embedding-table scale: Qwen’s table ($151,643 \times 3584$ parameters, ≈ 2.2 B weights) is $4.1\times$ larger than Mistral’s ($32,768 \times 4096$, ≈ 0.5 B) and $1.4\times$ larger than Llama’s ($128,000 \times 4096$, ≈ 1.6 B), contributing proportionally more total quantization error. All three models use Grouped-Query Attention, so attention head count is not the primary driver. The three-way rank consistency supports the vocabulary-sparsity hypothesis; a controlled ablation varying vocabulary while holding architecture fixed would be needed to confirm causality.

Downstream accuracy. To confirm that the +8.5% WikiText-2 PPL elevation does not translate to task degradation, we apply NLL-per-token scoring (no chat template) to both Qwen2.5-7B variants on ARC-Easy [Clark et al., 2018] ($n = 2,376$; 95% CI ± 2.0 pp). Table 8 presents the results. The q4 model scores +0.50pp above BF16—well within the confidence interval—confirming that the +8.5% perplexity penalty produces no statistically significant commonsense reasoning degradation at 7B scale.

Cross-corpus consistency (C4). The Δ PPL patterns observed on WikiText-2 generalise to C4 (web-crawled text). For Qwen2.5-1.5B, C4 yields Δ PPL of +16.2% vs. +15.0% on WikiText-2; for Qwen2.5-3B, +11.7% vs. +11.7%. The intra-family monotone ordering is preserved, and absolute deviations are within 2pp across the two corpora. This corpus-independence strengthens the interpretation of WikiText-2 Δ PPL as a model-intrinsic property rather than an evaluation artefact.

Mixed-precision failure. For the 1.5B dense

model, both mixed-precision recipes (3+6-bit and 2+6-bit) are *dominated* by uniform q4 on both size and quality axes. This result challenges the intuition that “less important layers can use fewer bits” for small dense models: the precision-sensitive layers at 1.5B scale appear distributed throughout the network, so that uniform quantization incurs less total rounding error than concentrated low-bit assignments under this configuration. Whether this holds across other small-scale architectures requires broader experimental validation.

Reproducibility (Contribution 2, enabling). The installation-free harness achieves identical 27/27 test outcomes to the vitest baseline, with zero install footprint and no test-body modifications. The migration cost is one import line per test file. This result confirms that Node.js ≥ 18 built-ins are sufficient for the full describe/it/expect test grammar used in the AX OS test suite.

Oracle switching (Contribution 3). The BRAIN_REAL=1 gate adds zero overhead in mock mode and generalises to any Python-backed JSON-serialisable oracle. The live path was executed once for end-to-end validation: the BRAIN API responded in 141.9s and returned `alpha_id RRd5Rvmj`, confirming the oracle gate is operational (see Section 5.2).

End-to-end coherence. Together, these three findings address the complete edge-deployment decision: the quantization measurements establish *which model and precision to deploy*; the harness provides *how to validate that decision reproducibly on the target machine*; and the oracle design pattern establishes *how to couple the deployed agent to its external simulator*. The Mistral-7B artifact is registered in the AEQIntegration registry, validated by the zero-dependency harness, and ready to load as a drop-in agent model—demonstrating all three components operating as an integrated system.

5.2 Limitations and Scope

Several important caveats apply to the findings.

Single hardware platform. All measurements are on Apple M1 Max with the MLX backend. Compression ratios should generalise (they reflect weight tensor sizes, which are hardware-independent), but

throughput figures will differ on CUDA GPUs. The RTX 5070 Ti (15.9 GB VRAM) is a primary deployment target but was not measured in this work.

Four Qwen scale points and two cross-architecture references. The intra-family finding (Δ PPL decreasing 1.5B to 7B, with the 14B result slightly above 7B) rests on four data points; measurements at 32B and above are needed to determine whether the sub-7B decrease resumes, plateaus, or the 14B uptick persists. The cross-architecture comparison involves two model pairs at the 7–8B scale: Mistral-7B (+4.4%) and Llama-3.1-8B (+6.9%) both outperform Qwen2.5-7B (+8.5%); other architectures (Phi, Gemma) remain uncharacterised. Extrapolating these trends below 1.5B or above 14B is a hypothesis, not an empirical result.

Single artifact per configuration. Each (model, precision) cell is a single quantized artifact evaluated once. Perplexity on a fixed corpus is deterministic given the artifact. MLX block-wise q4 quantization (RTN, group size 64) is a deterministic arithmetic operation on the BF16 weights: for each 64-weight block, `mx.quantize` computes the block’s absmx, maps weights to the nearest 4-bit integer, and stores the result—no calibration data and no PRNG calls are involved (verified by inspecting `mlx_lm.utils` source: zero `random` or `seed` calls in the quantization path). WikiText-2 evaluation on a fixed artifact is likewise deterministic on a fixed hardware platform. The reported Δ PPL values are therefore exact measurements, not run-to-run estimates. The relevant residual uncertainty is cross-library variance (e.g., GPTQ vs. MLX RTN at the same group size), which we identify as a direction for future work.

Quantization pipeline (uniform full-corpus). All six models in Table 2 are evaluated under an identical protocol: non-overlapping 512-token windows over the complete WikiText-2 test split, using each model’s native tokenizer. All Qwen q4 models were re-quantized locally with `mlx_lm.convert` at group size 64 (eliminating the prior mlx-community confound). Mistral-7B BF16 and q4 were evaluated under the same protocol. This uniform methodology enables a direct, protocol-matched cross-architecture comparison with no short-corpus caveats or inter-protocol uncertainty. The additional C4 evaluation (Section 4.4)

extends coverage to a second corpus; the BF16+q4 pair is fully measured for the two smallest models, and q4 absolute PPL is measured for the 7B models.

MoE models untested. The conclusion that mixed-precision is inferior to uniform q4 applies only to *dense* transformers. Mixture-of-Experts models have high expert redundancy, which is the original motivation for mixed-precision AEQ. This hypothesis remains untested.

BRAIN live simulation. The live oracle path (BRAIN_REAL=1) was executed once for validation: a test alpha expression was submitted to the WorldQuant BRAIN simulation API, which responded with a backtested performance record (alpha_id: RRd5Rvmj) after a measured latency of 141.9s—consistent with the architecture-derived estimate. A single validation run was performed to confirm end-to-end connectivity; broader oracle coupling experiments remain future work.

6 Conclusion

We presented AX OS as an integrated solution to three interdependent edge-deployment problems: selecting a model and precision level that retains acceptable quality under memory constraints; validating that selection reproducibly on the deployment machine without a build environment; and coupling the validated agent to an external simulator without compiled middleware. Addressing all three jointly, rather than in isolation, is the central contribution of this work.

The primary empirical finding is that, on WikiText-2 standard evaluation across six models (four Qwen2.5 sizes, Mistral-7B, and Llama-3.1-8B)—and confirmed on C4 web-crawled text for the two smallest models—uniform q4 Δ PPL within the Qwen family decreases from 1.5B to 7B (+15.0%, +11.7%, +8.5%) at $3.5\times$ compression (the 14B result is slightly above 7B; see Section 5.2). At the 7–8B scale, cross-architecture variation provides additional separation: Mistral-7B (+4.4%) and Llama-3.1-8B (+6.9%) are both more robust than Qwen2.5-7B (+8.5%), spanning a $1.9\times$ range and motivating architecture-aware rather than scale-only quantization scheme selection.

To make these measurements exactly repro-

ducible on any edge deployment target, we contribute two infrastructure components. The built-in-only `_harness.mjs` eliminates the vitest install requirement with a 40-line wrapper around Node.js built-ins, achieving 27/27 test pass—including the `registry-loop` tests that validate the compressed-model entries—on clean checkouts. The inline `BRAIN_REAL` oracle switching pattern couples a live Python simulation oracle to the agent demonstration script without requiring a compiled `dist/` directory, preserving portability across deployment platforms.

The Mistral-7B-Instruct-v0.3 q4 artifact (4,078 MB, ppl +4.4%, 35.6 tok/s on Apple M1 Max) is registered in the AX OS `AEQIntegration` registry as a drop-in agent model for consumer hardware with ≥ 5 GB free VRAM.

Future work includes extending the scale study to 32B and above parameter checkpoints to determine whether the sub-7B decreasing trend resumes, or the 14B uptick (+10.3% versus +8.5% at 7B) persists, validating mixed-precision AEQ on MoE architectures, and measuring throughput on CUDA targets to complement the Apple Silicon results. Having controlled for the quantization-pipeline confound (Section 5.2), both the intra-family scale trend and the cross-architecture gap now rest on a uniform full-corpus comparison.

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